

IDRIS SADIQ

Senior Data Engineer - Python, Spark, Airflow and Cloud

P +1 (424) 458-6548 E idrissdev@outlook.com IN linkedin.com/in/idris-s-3b49a23bb LOC Spring Valley, California

SUMMARY

Senior Data Engineer with 10+ years delivering enterprise SaaS, workflow automation, data-intensive services, and cloud-integrated products, specializing in Python 3.10-3.12, Spark 3.3-3.5, Airflow 2.6-2.10, and cloud warehouses on AWS and Azure; combines API and data architecture, legacy modernization, secure delivery, production troubleshooting, observability, and technical leadership to improve reliability, performance, release confidence, and measurable business outcomes.

EXPERIENCE

SumatoSoft

Senior Software Engineer | January 2021 - December 2025

- Architected customer solutions for **data platforms with Python, Spark, Airflow, Kafka, dbt, and cloud warehouses on AWS and Azure**, translating business processes into measurable use cases, governed data requirements, model interfaces, and production controls.
- Modernized notebooks and manual analysis into **curated datasets, dimensional models, governed data products, and replayable pipelines**, with versioned code, reproducible environments, automated validation, clear lineage, and documented operational ownership.
- Diagnosed production and evaluation failures involving **schema drift, late events, duplicate records, skewed joins, failed backfills, and freshness breaches**, using traceable datasets, offline benchmarks, online telemetry, failure-focused test cases, and controlled reprocessing.
- Implemented new ingestion, enrichment, retrieval, prediction, search, and human-review capabilities with secure APIs, asynchronous jobs, and explicit fallback behavior for uncertain outcomes.
- Established **data contracts, lineage, experiment tracking, prompt or model versioning, golden datasets, and release gates**, aligning research, engineering, product, and compliance expectations.
- Built CI/CD and MLOps automation for testing, packaging, deployment, rollback, scheduled evaluation, and cost monitoring across development, staging, and production environments.
- Strengthened privacy and security through access controls, secrets management, PII minimization, input filtering, audit logs, retention rules, and documented human-escalation procedures.
- Mentored engineers and analysts on production design, evaluation methodology, incident review, cost-aware model selection, data quality, and communication of limitations to stakeholders.
- Improved **pipeline runtime and warehouse spend by 36%** through targeted data-quality fixes, batching, caching, retrieval tuning, model optimization, partition changes, and removal of unnecessary processing stages.

High Touch Technologies

Senior Software Engineer | June 2018 - December 2020

- Developed analytical and automation capabilities for enterprise workflows using period-appropriate Python, cloud data services, secure integrations, and operational reporting technologies.
- Converted fragmented reports and manual data handling into scheduled pipelines, reusable services, governed schemas, and observable workflows with documented business definitions.
- Resolved defects involving **duplicate records, schema changes, stale aggregates, malformed partner data, slow transformations, and failed scheduled processing** across customer datasets.
- Implemented new enrichment, classification, routing, forecasting, search, and reporting capabilities with explainable outputs and backward-compatible interfaces for operational users.
- Improved consistency through deduplication, identity matching, event-time handling, validation rules, reconciliation jobs, and source-to-target checks across vendor and internal feeds.
- Expanded unit, data-quality, integration, performance, and regression testing with representative distributions, edge cases, and repeatable failure scenarios.
- Partnered with product, operations, QA, and customer teams to validate analytical behavior, investigate incidents, and document limitations before staged production releases.
- Reduced invalid or stale analytical outputs by **33%** through freshness checks, lineage, retry controls, reconciliation, schema governance, and monitored backfill procedures.

KEY ACHIEVEMENTS

Delivered Measurable Impact

Improved **pipeline runtime and warehouse spend by 36%** through targeted data-quality fixes, batching, caching, retrieval tuning, model optimization, partition changes, and removal of unnecessary processing stages.

Delivered Measurable Impact

Reduced invalid or stale analytical outputs by **33%** through freshness checks, lineage, retry controls, reconciliation, schema governance, and monitored backfill procedures.

Improved Service Reliability

Improved daily reporting reliability by **29%** through validation, deduplication, query tuning, job monitoring, and documented recovery steps for recurring data failure modes.

SKILLS

Data Engineering: Python 3.10-3.12, SQL, Spark 3.3-3.5, Airflow 2.6-2.10, dbt, Kafka, CDC, batch and streaming pipelines, data contracts, schema evolution

Cloud Data: AWS and Azure, object storage, managed Spark, serverless ETL, BigQuery/Snowflake/Redshift/Synapse, lakehouse patterns, orchestration, catalog and governance services

Modeling & Databases: dimensional modeling, Data Vault concepts, PostgreSQL, MySQL, SQL Server, MongoDB, Redis, partitioning, indexing, query optimization, slowly changing dimensions

Quality & Governance: Great Expectations/Deequ concepts, lineage, freshness, reconciliation, PII controls, RBAC, encryption, retention, observability, incident response, cost management

DevOps & Testing: Docker, Kubernetes, Terraform, GitHub Actions, unit and integration tests, data quality tests, backfill validation, contract testing, performance and load testing

Architecture: event-driven data platforms, lakehouse, medallion architecture, data products, metadata-driven pipelines, idempotency, replay, exactly-once tradeoffs, disaster recovery

Leadership: platform roadmaps, stakeholder alignment, mentoring, code and design review, data SLAs/SLOs, delivery planning, operational ownership, documentation

EXPERIENCE CONTINUED

Centric Tech Inc.

Software Engineer | September 2015 - May 2018

- Built data-processing and reporting features for business applications using Python, SQL, PostgreSQL, MySQL, SQL Server, scheduled jobs, and period-appropriate analytics tooling.
- Designed reusable extraction, validation, transformation, and export components that standardized data handling across customer modules and internal operational workflows.
- Fixed **incorrect totals, missing records, duplicate imports, date-boundary errors, slow aggregations, and inconsistent source identifiers** by tracing transformations end to end.
- Implemented new dashboards, reconciliation reports, search filters, notifications, and scheduled exports with explicit definitions and support-ready diagnostics.
- Improved query performance through targeted indexes, execution-plan analysis, incremental processing, partition-aware filters, and safer batch sizing for shared databases.
- Added structured logging, data-quality checks, retry limits, and audit trails so failed jobs could be diagnosed, replayed, and verified without uncontrolled duplicate processing.
- Improved daily reporting reliability by **29%** through validation, deduplication, query tuning, job monitoring, and documented recovery steps for recurring data failure modes.

Microsoft

Software Engineer Intern | September 2014 - August 2015

- Supported internal data and service improvements using period-appropriate **C#, .NET, SQL Server, Azure, JavaScript, and automated testing** under guidance from senior engineers.
- Built utilities that transformed and validated operational data, reducing repetitive manual handling while preserving traceable inputs, outputs, and error details for review.
- Improved diagnostics by adding contextual logs and validation messages around API, database, and background-processing paths that previously produced ambiguous failures.
- Assisted with integration changes by aligning payloads to established contracts, documenting edge cases, and preserving compatibility for dependent internal systems.
- Added unit and integration tests for data transformations, bug fixes, and service modules, incorporating review feedback on correctness, maintainability, and failure handling.
- Collaborated with engineers and product partners to clarify requirements, reproduce data discrepancies, validate fixes, and document support procedures for common issues.

SELECTED PROJECTS

Cloud Lakehouse and Streaming Platform

Designed batch and streaming pipelines with **Python, Spark, Airflow, Kafka, dbt, and AWS and Azure**, implementing schema evolution, deduplication, replay, lineage, quality checks, partition optimization, and governed data products for analytics and machine-learning consumers.

Operational Data Reliability Framework

Established data SLAs, freshness and volume monitoring, reconciliation, backfill controls, cost attribution, and incident runbooks; standardized reusable pipeline templates and contract tests so teams could deliver trustworthy datasets without repeating orchestration and governance logic.

PROFESSIONAL DEVELOPMENT

Certification details were not included in the supplied profile; list only credentials personally earned and currently valid. Role-aligned professional development includes **AWS and Azure data engineering, Databricks, Apache Airflow, Spark, streaming platforms, and cloud architecture**, supported by hands-on architecture, implementation, production support, mentoring, and security-focused engineering work.

EDUCATION

Harvard University | Bachelor's Degree, Computer Science | 2010 - 2014